AIDOOi

AID-LOG-003

BEDROOM COMPOSITION, FIT & BUDGET LOGIC

Revision 1 | MVP

Purpose: Define how AIDOOi converts Bedroom dimensions and the customer's preferred Bed Size into a complete initial furnishing composition, fit-aware product selection, room-level budget range, and dynamically recalculated final recommendation price.

1. Scope and Standard Role

AID-LOG-003 is the backend decision standard for the Bedroom customer path. It consumes Room Dimensions and Preferred Bed Size captured in AID-SPC-003, then uses AID-LOG-001 product price intelligence and the relevant category/data standards to construct the Bedroom recommendation.

The governing sequence is: **Room Dimensions -> Room Size -> Preferred Bed Size -> Composition -> Product Fit -> Budget -> Recommendation**.

2. Governing Principles

- The Bed is the anchor product. The customer's preferred Bed Size is fitted before the remaining bedroom composition is finalized.
- Length x Width determines Bedroom Area and the system Room Size Classification; the customer does not manually select the size class.
- Bed Size has priority over a generic size-based composition. Two bedrooms with the same area may receive different compositions when their selected Bed Sizes differ.
- After fitting the Bed, AIDOOi evaluates remaining usable space, circulation and furniture access before adding the rest of the composition.
- AIDOOi initially aims to generate a complete Bedroom composition, but products may be reduced or removed when real fit constraints require it.
- Default Composition is not a hard constraint, and Preferred Dimensions are not absolute cut-off filters.
- Fit and Circulation take priority over Budget Preference. A product that does not fit must not be recommended merely to satisfy a budget tier.
- After recommendation, the customer may remove products and the Final Composition Price must recalculate dynamically.

3. Bedroom Size Classification

Calculated Area	Classification
< 9 m²	Compact
9-14 m²	Small
14-20 m²	Medium
20-30 m²	Large
30+ m²	Extra Large

Calculation: **Length x Width = Bedroom Area**. Room Size activates a default profile, but Preferred Bed Size may modify the resulting composition.

4. Bed-as-Anchor and Fit Priority

The system must first fit the selected Bed Size, then preserve usable circulation and access before filling secondary furniture slots. Bedroom fit priority is generally:

Bed -> Circulation -> Bedside Tables -> Wardrobe -> Chest of Drawers -> Rug / Lighting -> Seating -> Decor.

Wardrobe fit must consider not only the product width/depth but also door opening or sliding access and usable circulation. When a selected Bed Size consumes more space than the default profile anticipates, secondary products may be reduced in quantity, downsized, or omitted.

5. Size-Based Default Composition and Preferred Fit

5.1 Compact Bedroom - < 9 m²

Product	Default Qty	Preferred Fit Logic
Bed	1	Customer-selected Bed Size; anchor
Bedside Table	1	Compact; approx. 30-40 cm width
Wardrobe	1	2-Door / Small; approx. 80-120 cm width
Rug	1	Small; based on remaining usable space
Table / Wall Light	1	Space-efficient; wall light permitted
Mirror	1	Prefer wall-mounted

Product	Default Qty	Preferred Fit Logic
Decor	1-2	Primarily wall/surface decor

5.2 Small Bedroom - 9-14 m²

Product	Default Qty	Preferred Fit Logic
Bed	1	Customer-selected Bed Size; anchor
Bedside Table	2	Approx. 35-50 cm width; may reduce to 1
Wardrobe	1	2-3 Door; approx. 100-160 cm width
Chest of Drawers	1	Compact; approx. 60-100 cm width
Rug	1	Small / Medium; fit to Bed and free space
Table / Wall Light	2	Preferably one on each side of Bed
Mirror	1	Wall-mounted or compact floor mirror
Decor	2-3	Placement-zone driven

5.3 Medium Bedroom - 14-20 m²

Product	Default Qty	Preferred Fit Logic
Bed	1	Customer-selected Bed Size; anchor
Bedside Table	2	Approx. 40-55 cm width
Wardrobe	1	3-4 Door / Medium; approx. 140-220 cm
Chest of Drawers	1	Approx. 80-120 cm
Accent Chair / Armchair	1	Compact; approx. 60-80 cm width
Rug	1	Medium / Large; matched to Bed Size
Table / Wall Light	2	One on each side of Bed
Mirror	1	Wall or floor mirror
Decor	3-4	Placement-zone driven

5.4 Large Bedroom - 20-30 m²

Product	Default Qty	Preferred Fit Logic
Bed	1	Customer-selected Bed Size; anchor
Bedside Table	2	Approx. 45-60 cm width
Wardrobe	1	4-5 Door / Medium-Large; approx. 180-280 cm
Chest of Drawers	1	Approx. 100-140 cm
Accent Chair / Armchair	1	Approx. 70-90 cm width
Side Table	1	Approx. 35-50 cm
Rug	1	Large / XL; matched to Bed and layout
Table / Wall Light	2	Bedside lighting
Floor Lamp	1	If Reading/Seating Zone fits
Mirror	1	Floor or wall mirror
Decor	4-5	Placement-zone driven

5.5 Extra Large Bedroom - 30+ m²

Product	Default Qty	Preferred Fit Logic
Bed	1	Customer-selected Bed Size; anchor
Bedside Table	2	Approx. 50-70 cm width
Wardrobe	1	5+ Door / Large; approx. 240-350+ cm

Product	Default Qty	Preferred Fit Logic
Chest of Drawers	1	Approx. 110-160 cm
Accent Chair / Armchair	2	Approx. 70-95 cm each
Side Table	1	Approx. 40-60 cm
Rug	1	XL; matched to Bed and layout
Table / Wall Light	2	Bedside lighting
Floor Lamp	1	For Seating / Reading Zone
Mirror	1	Large floor or wall mirror
Decor	5-7	Placement-zone driven

6. Bedroom Placement and Secondary Zone Logic

Decor must be selected according to usable placement zones rather than a fixed list. Wall zones may receive Wall Art or Mirror; furniture surfaces may receive Vase, Decorative Object or Candle Holder; open floor/corner zones may receive a Planter or other suitable floor decor.

For Large Bedrooms, an Accent Chair + Side Table + optional Floor Lamp may form a small Reading/Seating Zone when the layout supports it.

For Extra Large Bedrooms, AIDOOi may create a stronger Secondary Seating/Reading Zone, typically using 2 Accent Chairs + 1 Side Table + 1 Floor Lamp. This zone is not mandatory and must not be added merely to fill space or increase spend.

7. Bedroom Budget Tiers

Room Size	Essential	Value	Standard	Premium	Luxury
Compact <9	1,000-2,000	2,000-3,500	3,500-6,000	6,000-10,000	10,000+
Small 9-14	1,500-3,000	3,000-5,000	5,000-8,000	8,000-14,000	14,000+
Medium 14-20	2,500-4,500	4,500-7,000	7,000-11,000	11,000-18,000	18,000+
Large 20-30	3,500-6,000	6,000-9,500	9,500-15,000	15,000-25,000	25,000+
Extra Large 30+	5,000-8,000	8,000-13,000	13,000-20,000	20,000-33,000	33,000+

All ranges are AED and represent the target budget for the initial complete Bedroom composition. Bed Size is handled within allocation and fit rather than creating a second customer-facing budget table.

8. Budget Allocation Logic

- The customer selects one Bedroom-level tier: Essential, Value, Standard, Premium or Luxury.
- The applicable Room Size activates the total target range.
- Bed Size influences the internal allocation because the Bed is the primary fit anchor and an important cost anchor.
- Bed and Wardrobe normally represent the principal budget anchors in a complete Bedroom.
- AIDOOi may mix product-level tiers from AID-LOG-001. A Standard Bedroom does not require every selected product to be Standard.
- If a larger Bed consumes more of the selected room budget, secondary items may be sourced from lower product tiers while preserving fit, style, colour and usability.
- If no valid composition satisfies both fit and the selected budget, fit must not be sacrificed. The system should identify the constraint rather than recommend an unsuitable product.

9. Selected Room Budget vs Final Composition Price

Selected Bedroom Budget Tier is the target used to assemble the initial recommendation. **Final Composition Price** is the sum of Current Selling Prices of the products that remain selected.

The customer may remove items after viewing the complete Bedroom recommendation. The system must recalculate the total immediately. The revised price is not required to remain within the originally selected Bedroom tier.

Formula: Final Composition Price = Sum of Current Selling Price of all selected/remaining product offers.

10. Recommendation Sequence

Step	System Action
1	Receive Bedroom Length, Width and Preferred Bed Size from AID-SPC-003.
2	Calculate area and assign Compact / Small / Medium / Large / Extra Large.
3	Fit the selected Bed Size as the anchor product.
4	Evaluate remaining usable space, circulation and wardrobe/access requirements.
5	Load and adapt the default composition and preferred-fit profile.
6	Load the five applicable Bedroom budget ranges and receive the customer's tier selection.
7	Use AID-LOG-001 plus Style, Colour and availability rules to assemble candidate compositions.
8	Rank and present a complete fit-valid Bedroom recommendation.
9	Allow product removal and dynamically recalculate Final Composition Price.

11. Relationship to Other Standards

Reference	Role
AID-SPC-003	Bedroom Path - captures Room Dimensions, Preferred Bed Size, optional photos/floor plan, Style, Colour and Budget journey.
AID-LOG-001	Product Price Range Standard - individual product price intelligence.
AID-CAT-004	Beds Standard - Bed Size and bed-specific practical attributes.
AID-CAT-005	Storage & Casegoods - Wardrobe, Chest of Drawers and Bedside Table taxonomy/attributes.
AID-DATA-007	Product Colour Standard.
AID-DATA-009	Product Style Standard.
AID-DATA-010	Product Pricing Standard - Current Selling Price and offer-level price handling.
AID-DATA-011	Product Availability & Inventory Standard.

12. Governance and MVP Boundaries

- The size thresholds, quantities, preferred dimensions and Bedroom budget ranges in this revision are approved MVP configuration values.
- Preferred dimensions are guidance, not hard filters; actual layout fit is authoritative.
- The customer's selected Bed Size must not be silently changed by the engine merely to preserve a default composition.
- If the preferred Bed cannot be accommodated with acceptable fit/circulation, the system should surface the constraint rather than pretend the room is valid.
- Future revisions may refine thresholds, fit rules and ranges using live inventory, layout performance, conversion data and customer behavior.
- Dining Room, Home Office, Studio Apartment, Full Apartment and Villa logic remain separate LOG standards.

13. Standard Summary

AID-LOG-003 converts Bedroom dimensions and the customer's Preferred Bed Size into a fit-aware complete initial Bedroom recommendation. The Bed is fitted first as the anchor; remaining usable space, circulation and access then determine the rest of the composition. A size-specific total budget range guides product selection using AID-LOG-001, while product-tier mixing is permitted. After recommendation, customer removals dynamically update the Final Composition Price.